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# Exploration of oral microbiota alteration and AI-driven non-invasive hyperspectral imaging for CAD prediction

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## Abstract

**Background** Oral microbiome dysbiosis is an important risk factor affecting the occurrence and progression of coronary artery disease (CAD). However, the dysbiosis on the tongue in patients with CAD is still unclear, and whether the oral alteration caused by these disorders can be identified by other tools for CAD diagnosis needs to be further explored. Hyperspectral imaging (HSI) is characterized as high spectral resolution, broad spectral range, and superior spatial resolution. Hyperspectral images contain high-dimensional data that generally require machine learning algorithms for feature identification and model construction. Therefore, this study aims to investigate the variation of tongue microbiota and the effectiveness of HSI models in CAD diagnosis.

**Methods** Between 2023 and 2024, we prospectively approached 276 patients with chest pain and exhibiting risk for CAD who underwent coronary artery angiography (CAG). And 190 patients were enrolled in this study. Tongue dorsum swabs were collected for subsequent 16sRNA sequencing and microbiome analysis. Tongue dorsum features were extracted from hyperspectral images. The HSI analysis incorporated a total of 4750 hyperspectral images from all patients. All images are divided into training set ( $N=2555$ ), internal test set ( $N=1095$ ) and external test set ( $N=1095$ ). A total of 31 models were constructed. 30 single machine learning algorithms were used to construct and test the CAD prediction models. Furthermore, the best performing fusion model was established. The efficacy of the model was evaluated employing several metrics, including area under the curve (AUC), decision curve analysis (DCA), calibration curve, accuracy (ACC), sensitivity (SE), specificity (SP), positive predictive value (PPV), negative predictive value (NPV) and F1 score.

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**Results** The 16sRNA sequencing results indicated significant dysbiosis in the oral microbiota of patients with CAD, with decreased microbial abundance, network complexity and stability. The fusion model (GP-GB-SVM) demonstrated the highest performance, achieving an AUC of 0.92, ACC of 0.82, SE of 0.70, SP of 0.92, PPV of 0.88 and NPV of 0.79 in the internal test set and AUC of 0.86, ACC of 0.70, SE of 0.90, SP of 0.46, PPV of 0.60 and NPV of 0.90 in the external test set.

**Conclusion** These findings not only emphasize the significant alteration of microbiome colonized on the tongue dorsum in CAD patients but also demonstrate the tongue features associated with microbiome dysbiosis can be identified in hyperspectral images. Thereby the integration of HSI and machine learning provides novel insights into non-invasive diagnosis of CAD.

**Keywords** Coronary artery disease, Microbiome, Hyperspectral imaging, Machine learning, Prediction models

## Introduction

Coronary artery disease (CAD) remains the leading cause of mortality and chronic disability within the spectrum of cardiovascular diseases worldwide [1]. Recent studies have increasingly highlighted the close relationship between oral health and CAD [2, 3]. Dysbiosis within the oral microbiota constitutes a significant contributing factor to the pathogenesis of oral diseases, a significantly increased risk of cardiovascular disease and adverse cardiovascular events among those with poor oral health [4]. Previous research has substantiated alterations in the oral microbiota associated with diseases such as diabetes and CAD [5]. Additionally, the presence of common pathogenic bacteria from the oral cavity was identified in arterial plaques and within the arterial intima [6–8]. These oral pathogens facilitate the development and advancement of atherosclerotic plaques by mechanisms including the induction of immune-mediated inflammation and the disruption of metabolic homeostasis [2]. The dorsal tongue harbors a substantial microbial community and is integral to oral health [9]. However, it is unclear whether alterations in the microbiota colonized on the dorsal tongue are associated with CAD. Therefore, it is necessary to explore more alterations of microbiota on the tongue dorsum to further clarify the oral microbiome dysbiosis in CAD patients.

Several facial features, such as alopecia, earlobe crease, and xanthelasmata, have long been associated with an increased risk of CAD [10, 11]. Additionally, the integration of facial imaging with deep learning algorithms has proven to be an effective method for screening patients with CAD [12, 13]. These studies suggest that superficial characteristics may serve as non-invasive predictors of CAD risk. However, previous studies have predominantly focused on alterations in facial skin characteristics, which may be subject to the influence of diverse ethnic skin tones. Notably, oral hyperspectral imaging (HSI) constitutes a high-precision classification technique by offering comprehensive spectral information for each pixel within an image [14]. Thus, oral HSI enables a thorough characterization and enhanced sensing perspective of target

objects by utilizing its high spectral resolution, broad spectral range, and superior spatial resolution [15–18]. Previous studies have confirmed that HSI enables disease diagnosis primarily by identifying characteristic spectra reflected from the body surface or the tissue, such as the diagnosis of diabetic foot by characteristic spectral alterations of hemoglobin on the skin surface. Therefore, large discrepancies in skin color on the body surface may have implications for the recognition of characteristic spectra [19, 20]. The utilization of oral HSI could effectively mitigate the impact of varying skin tones on superficial skin features. Additionally, modifications in the microbiota of the dorsal tongue exert an influence on its morphology, moisture levels, coloration, and other characteristics [21]. These alterations can be assessed through dorsal tongue imaging techniques. Therefore, harnessing the advantages of oral HSI can facilitate the identification of additional surface characteristics relevant to the alteration of microbiota on the tongue dorsum.

Therefore, we posited that significant alteration of microbiota on the dorsal tongue can be observed in CAD patients and surface characteristics on the dorsal tongue can be reflected in hyperspectral images. This study aimed to clarify the variations of microbiota on the dorsal tongue in CAD patients and conduct multiple machine learning algorithms to acquire and incorporate tongue features for detecting CAD.

## Methods

### Study design and participants

The study consisted of two phases. In phase one, we prospectively enrolled 211 chest pain patients in Renmin Hospital of Wuhan University to develop and test the CAD models (between July 2023 and August 2023). In phase two, 65 chest pain patients were enrolled in Yichang Central People's Hospital to test the established models in phase one (between August 2024 and October 2024). Coronary angiography (CAG) was performed in patients. CAD patients were defined as at least one coronary artery with stenosis of  $\geq 50\%$  in CAG. The exclusion criteria were as follows: (1) history of previous

tongue surgery; (2) clinical baseline characteristics missing > 30%; (3) refusal of hyperspectral image capture; (4) acute myocardial infarction. Following the application of research criteria, 86 individuals were excluded, leaving 190 eligible participants. All eligible participants signed informed consent for study involvement, including voluntarism, information disclosure, decision-making capacity, and use of their hyperspectral images and swab of tongue in research before the CAG procedure. The research protocol and consent processes received approval from the Institutional Review Board of Renmin Hospital of Wuhan University (WDRY2023-K084, chict.org.cn identifiers ChiCTR2300073683, Registration Date: 2023-7-19) and Yichang Central People's Hospital (No. 2024-216-01). The procedures in this study adhered to the tenets of the Declaration of Helsinki.

### Specimen collection and 16sRNA gene sequence

The participants were divided into CAD group and control group based on the results of CAG. Tongue coating specimens were obtained in the morning before participants' dental hygiene routines and food consumption. A tongue scraper was utilized to collect samples from the posterior to anterior regions of the tongue, repeating the process 5–7 times. The gathered coatings were placed in a 1.5 ml Eppendorf vessel containing phosphate-buffered saline solution. Subsequently, the tubes underwent centrifugation at 5000 rpm for 10 min. The resulting supernatant was discarded, while the precipitate was retained and preserved at -80 °C until analysis. Bacterial DNA extraction was performed for PCR amplification using specific primers with barcodes targeting distinct regions (16SV3-V4) of 16sRNA genes. All PCR reactions were conducted using 15 µL of Phusion® High-Fidelity PCR Master Mix (New England Biolabs), 0.2 µM of forward and reverse primers, and approximately 10 ng of template DNA. The thermal cycling protocol consisted of initial denaturation at 98 °C for 1 min, followed by 30 cycles of denaturation at 98 °C for 10 s, annealing at 50 °C for 30 s, and elongation at 72 °C for 30 s, with a final extension at 72 °C for 5 min. DNA libraries were prepared per the supplier's protocol, and DNA sequencing was subsequently performed.

### Sequence assembly and analysis

The tongue coat microbiome's 16sRNA sequence information from participants was utilized for comparative analysis. Sequence compilation and examination were conducted. In brief, paired-end reads were allocated to samples according to their distinct barcode and truncated by removing the barcode and primer sequence. FLASH (V1.2.11, <http://ccb.jhu.edu/software/FLASH/>) was employed to merge paired-end reads. Raw tags underwent quality filtration using fastp (Version 0.23.1)

software to yield high-quality Clean Tags. The tags were then evaluated against the reference database [Silva database 16 S, <https://www.arb-silva.de/> Unite Database (ITS), <https://unite.ut.ee/>] utilizing the UCHIME Algorithm ([http://www.drive5.com/usearch/manual/uchime\\_algo.html](http://www.drive5.com/usearch/manual/uchime_algo.html)) to identify chimera sequences, which were subsequently eliminated.

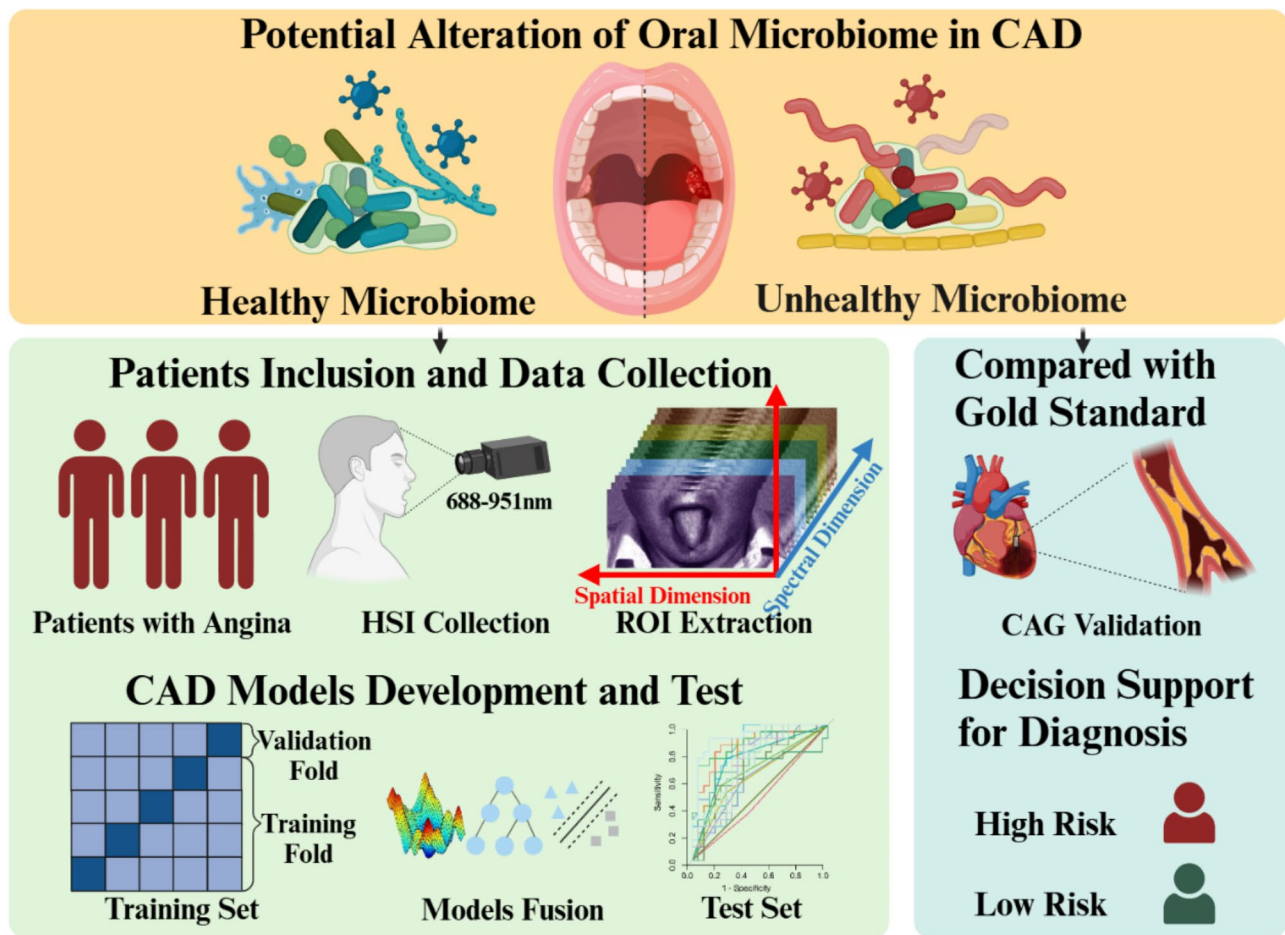
Alpha diversity was computed utilizing 4 metrics in QIIME2, encompassing Chao1, Shannon, Simpson and Pielou\_e to examine the diversity, richness and evenness of the communities across groups. Beta diversity was determined based on weighted and unweighted unifracs distances in QIIME2 to assess the intricacy of the community composition and contrast the disparities between CAD and control groups. Principal coordinate analysis (PCoA) was employed to elucidate the variation in microbial communities between two groups utilizing weighted unifracs values. T-tests were conducted to detect differential bacteria at phylum and genus levels between the two groups. Linear discriminant analysis (LDA) effect size (LEfSe) was utilized to pinpoint key bacteria in tongue coating samples between two groups at multiple levels in datasets. The key bacteria were ranked according to the outcomes of a Mann–Whitney U-test, which ascertained the features with significantly distinct abundances between assigned taxa and employed LDA to evaluate the effect size of each feature and visualize the findings using taxonomic bar charts and cladograms. The P values were adjusted as outlined by Benjamini and Hochberg. Network analysis was used to explore the symbiotic relationship between different bacterium genus and to reveal the two groups community structures.

### Hyperspectral image study setting

Every participant contributed 25 qualified HSI images, forming the final analysis dataset. Totally 4750 images were collected to develop and test the models. All eligible images in phase one were recruited and randomly assigned to training (70%,  $N=2555$ ) and internal test set (30%,  $N=1095$ ). The images in phase two were utilized as the external test (30%,  $N=1095$ ) to evaluate the established models. HSI of the tongue dorsum were captured in all eligible patients before CAG. Baseline characteristics and results of CAG were collected, while features extracted from hyperspectral images were combined with machine learning algorithms to develop prediction models. Two interventional cardiologists, blinded to the research protocol and patient data, independently evaluated CAG outcomes to determine the presence of CAD abnormalities (Fig. 1).

### Hyperspectral image collection

Hyperspectral imaging was performed using the XIMEA USB3 Vision Camera (XIMEA Inc., Münster, German).



**Fig. 1** Oral microbiome alteration in CAD patients and application of oral hyperspectral image for CAD diagnosis: An overview

Before image acquisition, patients were instructed to refrain from eating and to rinse their mouths with sterilized saline. The camera captured 25 spectral bands within the 688–951 nm wavelength range, featuring a spectral resolution ranging from 4 to 25 nm and a maximum spatial resolution of  $644 \times 484$  pixels. A patient can be captured 25 images in 25 different spectral bands in a single shot. An illumination setup utilizing a 150 W quartz tungsten halogen light source was implemented. The distance between the camera's optics and the upper surface of the tongue was 100 cm. The total image acquisition time per patient was less than one second, and this rapid acquisition was generally well tolerated by the patients. In instances where tongue movement interfered with image acquisition, repeat imaging was conducted.

#### Pre-processing and annotation of hyperspectral image data

The raw hyperspectral images were subjected to initial processing to mitigate the impact of ambient light conditions and sensor dark currents, following the previously described procedure [22]. The raw images were

calibrated and standardized prior to subsequent analysis. The spectral and spatial features were extracted across 25 bands. The preprocessing and analysis of the HSI data were performed in ENVI5.3 (The environment for visualizing images, ENVI, version 5.3, USA). This study centers on the prediction of CAD, as evidenced by CAG findings. Specifically, CAD is characterized as at least one coronary artery with stenosis of  $\geq 50\%$  on coronary angiography [23].

#### Hyperspectral features and models construction

A total of 30 single machine learning algorithms were employed to train predictive models for predicting CAD, including 16 Linear models (Theil-Sen estimator, Hube, etc.), 5 Nonlinear models [Gaussian Process (GP), Support Vector Machine (SVM), etc.], 4 Bayesian models (Automatic Relevance Determination, Bernoulli Naive Bayes, etc.), 3 Decision Trees and Ensemble models (Gradient Boosting (GB), Random Forest (RF), etc), 2 Neural Networks (Perceptron and Multilayer Perceptron). The CAD prediction models were developed by utilizing spectral and spatial features across 25 bands for each



patient in the training set, which encompassed characteristics of the tongue dorsum. Random Forest (RF) was used in the training set for selecting the most significant HSI features to develop the models. This involves selecting appropriate hyperparameters and fitting the model in the training data. Subsequently, the feature importance scores were calculated and the features were ranked based on their importance scores. According to the importance scores, these features were applied to model construction until the performance of the model did not improve. To develop prediction models, fivefold cross-validation was performed in the training set with random reordering. To implement a five-fold cross-validation, the training set dataset was first randomly partitioned into five subsets. The models were trained on four of these subsets and validated on the remaining subset. This process is repeated five times, with each subset serving as the validation set once. This method helps mitigate overfitting and ensures that the model generalizes well to unseen data. The prediction accuracy was determined by comparing the model outputs with the established ground truth based on CAG results.

#### Prediction model upgradation

To improve the performance of CAD prediction models, a decision-level fusion strategy was used to achieve model fusion. 30 single models were trained separately on different feature sets in the training set. Each model outputs its prediction scores. Their predictions were aggregated at the decision level based on weighted voting. Specifically, all models were unduplicated and randomly combined in pairs, three by three until all individual models were entirely combined, for a total of 1,073,741,793 combinations. Finally, the best combination of all fusion models and individual models was selected by ROC and F1 score comparison in both external and internal test sets.

#### Diagnostic performance of the prediction model

To assess the performance of the CAD prediction models, we calculated the sensitivity (SE), specificity (SP), area under the receiver operating characteristic curve (AUC), decision curve analysis (DCA), calibration curve, positive predictive value (PPV), negative predictive value (NPV), accuracy (ACC) and F1 score.

#### Statistical analysis

Data are displayed as average  $\pm$  standard deviation for continuous variables and proportions for categorical variables. Comparisons of continuous data were conducted using one-way ANOVA or Kruskal-Wallis rank sum test, while categorical data were evaluated using chi-square or Fisher's exact tests. All analyses were two-tailed, with statistical significance set at  $P < 0.05$ . Statistical analyses

were conducted using SPSS Statistics version 29 and Python version 3.12.

#### Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

#### Results

##### Oral dysbiosis in CAD patients

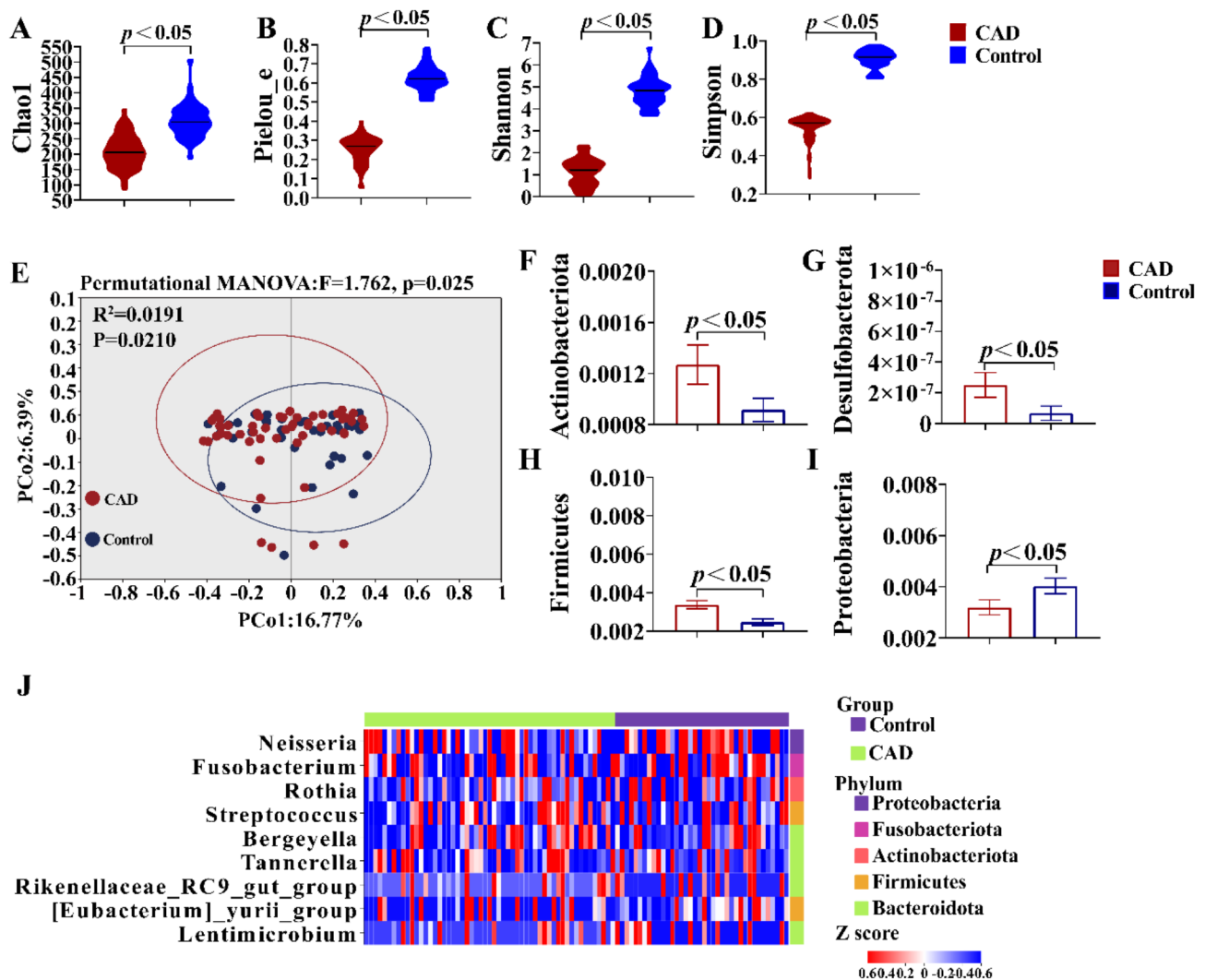
The swabs of the participants were analyzed using 16sRNA gene sequencing. The  $\alpha$  diversity of the oral microbiota in CAD was significantly altered, suggesting oral dysbiosis (Fig. 2A-D). The community diversity and richness indexes, including Chao1, Pielou\_e, Shannon and Simpson, showed a significant decrease in the CAD group compared to the control group. PCoA analyses revealed that there was a significant difference between the CAD group and the control group ( $P < 0.05$ ) (Fig. 2E).

Analysis at the phylum level showed that CAD patients presented significantly higher relative abundance of *Actinobacteria*, *Firmicutes* and *Desulfobacterota* (all  $P < 0.05$ ), alongside a significantly lower relative abundance of *Proteobacteria* ( $P < 0.001$ ) (Fig. 2F-I). Of the nine discriminatory genera, the relative abundance of *Neisseria* and *Eubacterium\_yurii\_group* were significantly higher in the control group, and the others were significantly higher in the CAD group, including *Streptococcus*, *Rothia*, *Fusobacterium*, *Tannerella*, *Bergeyella*, *Lentimicrobium* and *Rikenellaceae\_RC9\_gut\_group* (Fig. 2J).

LEfSe analysis was performed to identify the discriminatory taxa in the tongue coating microbiota between the two groups (Supplemental Fig. 1A-B). The CAD patient was characterized by a preponderance of *p\_Firmicutes*, *o\_Lactobacillales*, *c\_Bacilli*, *f\_Streptococcaceae*, *g\_Streptococcus*, *o\_Coriobacteriales*, *c\_Coriobacteriia*, *f\_Atopobiaceae* and *s\_Actinomyces\_genomosp\_P1\_oral\_clone\_MB6\_C03* [LDA score ( $\log_{10}$ )  $> 3$ ], whereas the control group was characterized by a preponderance of *s\_Prevotellaceae\_bacterium\_Marseille\_P2826*, *s\_Prevotella\_shahii*, *o\_Sphingomonadales*, *g\_Alloprevotella*, *s\_Haemophilus\_parainfluenzae*, *g\_Fusobacterium*, *f\_Fusobacteriaceae*, *s\_Fusobacterium\_periodonticum*, *o\_Burkholderiales*, *g\_Neisseria*, *f\_Neisseriaceae*, *c\_Gammaproteobacteria*, and *p\_Proteobacteria* [LDA score ( $\log_{10}$ )  $> 3$ ].

##### The reconstruction of the oral microbiota in CAD

Network analysis was conducted to evaluate the correlation of the oral microbiota, with the aim of further elucidating the characteristics of the microbiota in patients with CAD. 13 genera of microbiota demonstrated significant associations between the control group and CAD group (Fig. 3A). The complexity of the network exhibited

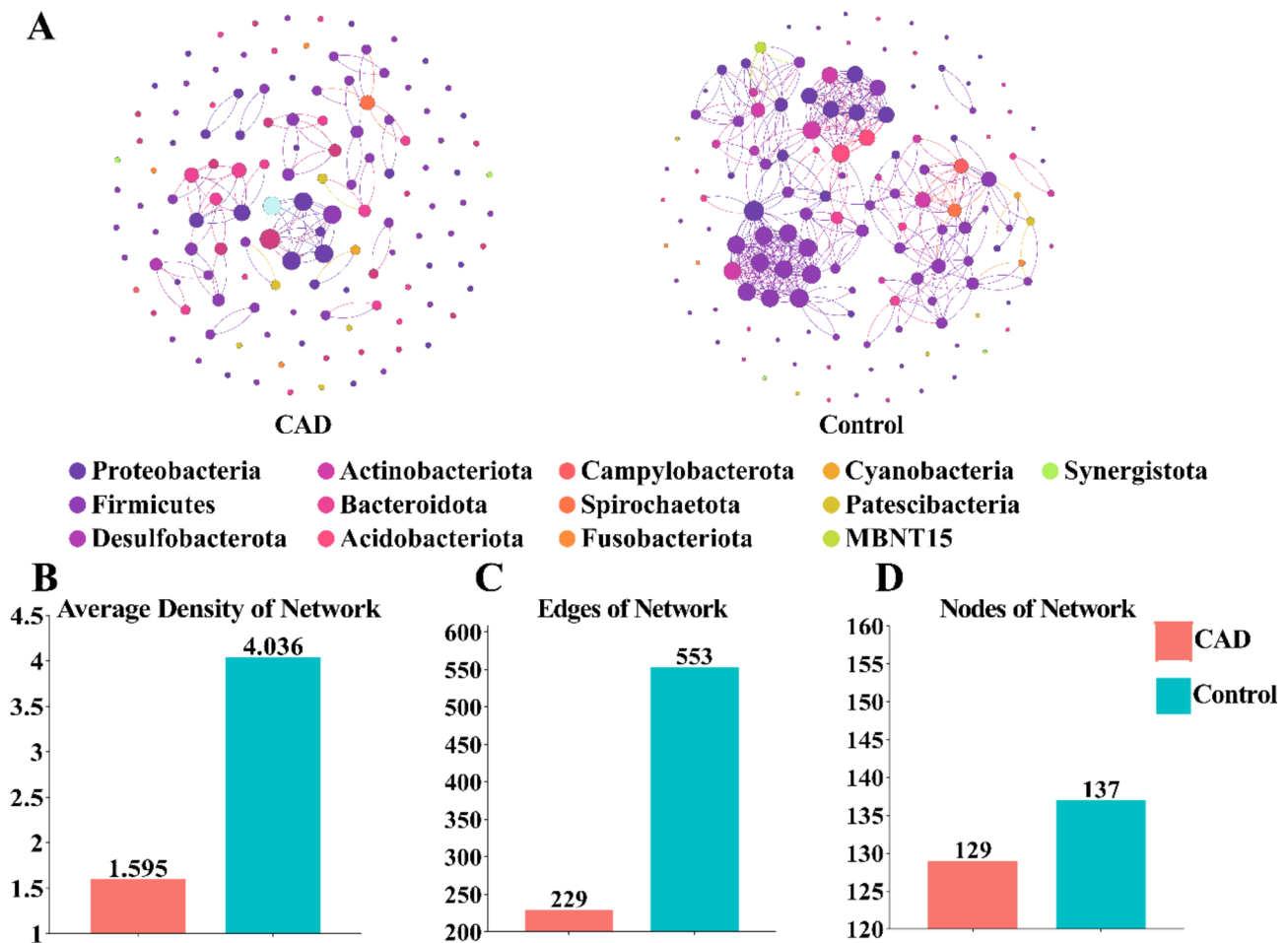


**Fig. 2** Microbial Dysbiosis in CAD Patients. **A-D**. Comparison of  $\alpha$ -diversity. Comparison of  $\alpha$  diversity of the oral microbiota in CAD patients and control patients. Compared with the control group, patients in the CAD group had a significantly lower alpha diversity index. **E**. PCoA analysis. Plot of principal coordinates analysis (PCoA) based on the Bray–Curtis distance matrix of the 16 sRNA gene sequencing data. According to the Bray–Curtis distance analysis, it can be visualized that oral microbiota differed significantly between the two groups of participants. **F-I**. Comparison of phylum levels. Comparison of the relative abundances of oral bacterial the CAD and control groups at the genus level. **J**. Comparison of genus levels. Heatmap of the relative abundances of the 9 significantly differentially bacteria in the CAD group compared with the control group at the genus level

significant differences between the two groups. Specifically, the network analysis revealed that the topological properties, encompassing the number of nodes, edges, and average degree, were markedly higher in the control group. Quantitatively, the comparison between the 2 groups demonstrated the following: nodes (129 vs. 137), edges (229 vs. 553), and average density (1.595 vs. 4.036) (Fig. 3B-D). The presence of elevated topological properties in control individuals suggests a heightened complexity and robustness within the microbial network structure.

#### Model development and HSI features exhibition

276 patients with chest pain were prospectively approached and underwent coronary artery angiography (CAG). After screening according to inclusion and exclusion criteria 190 patients were enrolled in this study. Our dataset consisted of 190 participants from two centers, accompanied by a total of 4,750 HSI images. In phase one, 3650 eligible images from Renmin Hospital of Wuhan University were randomly allocated into two groups: 70% ( $n=2555$ ) were assigned to the training set, and 30% ( $n=1095$ ) were assigned to the internal test set. In phase two, 1095 eligible images from Yichang Central People's Hospital were enrolled in an external test set



**Fig. 3** Microbial Network Reconstruction in CAD Patients. **A.** Network Correlation Analysis. Through network analysis, 13 genera of oral microbiota at the genus level exhibited potential correlations. **B-D.** Topological properties of network analysis. Comparison of topological properties between CAD and control groups. These results indicated that the microbiota network of the control group was relatively more stable than the control group

(Supplemental Fig. 2). The baseline characteristics were similar between the training and test sets. (Table 1)

The training set was utilized for feature selection and model development, employing various machine learning algorithms with five-fold cross-validation. Furthermore, based on decision level fusion strategy individual model was integrated to optimize overall model performance. The predictive efficacy of these models was subsequently evaluated using the CAG imaging of the participants within the test set. Following the identification and ranking of tongue dorsum HSI features utilizing random forests, 30 individual prediction models were developed employing various machine learning algorithms. To further improve model performance, a decision level fusion strategy based on the AUC was utilized to integrate individual models. The best performing AUC combination identified was GP-GB-SVM (Fig. 4A). The analysis of spectral and texture features revealed significant disparities in the spectral and textural characteristics between the control and CAD patients (Fig. 4B).

#### Model evaluation in internal and external test sets

Totally 31 distinct machine learning algorithms were used for model development. 30 individual models were constructed and 1 fusion model was established based on decision level strategy. The six highest-performing models, as ranked by the AUC, are GP-GB-SVM (fusion model), GP, GB, SVM, RF, and TheilSen. The AUC values of the aforementioned six models in the internal test set range from 0.783 to 0.921 [GP-GB-SVM:0.921(0.839–1), GP:0.879(0.779–0.980), GB:0.854(0.737–0.972), SVM:0.844(0.728–0.959), RF:0.827(0.704–0.950), TheilSen:0.783(0.646–0.920)] and in the external test set range from 0.685 to 0.860 [GP-GB-SVM:0.860(0.754–0.967), GP:0.841(0.723–0.964), GB:0.844(0.728–0.959), SVM:0.744(0.596–0.892), RF:0.811(0.685–0.938), TheilSen:0.685(0.525–0.846)] (Fig. 5A and Fig. 5D). The calibration curve demonstrated that the diagnostic performance of GP-GB-SVM, GP, GB, and RF models was satisfactory in the internal test set; however, they exhibited a tendency to overestimate

**Table 1** Baseline characteristics

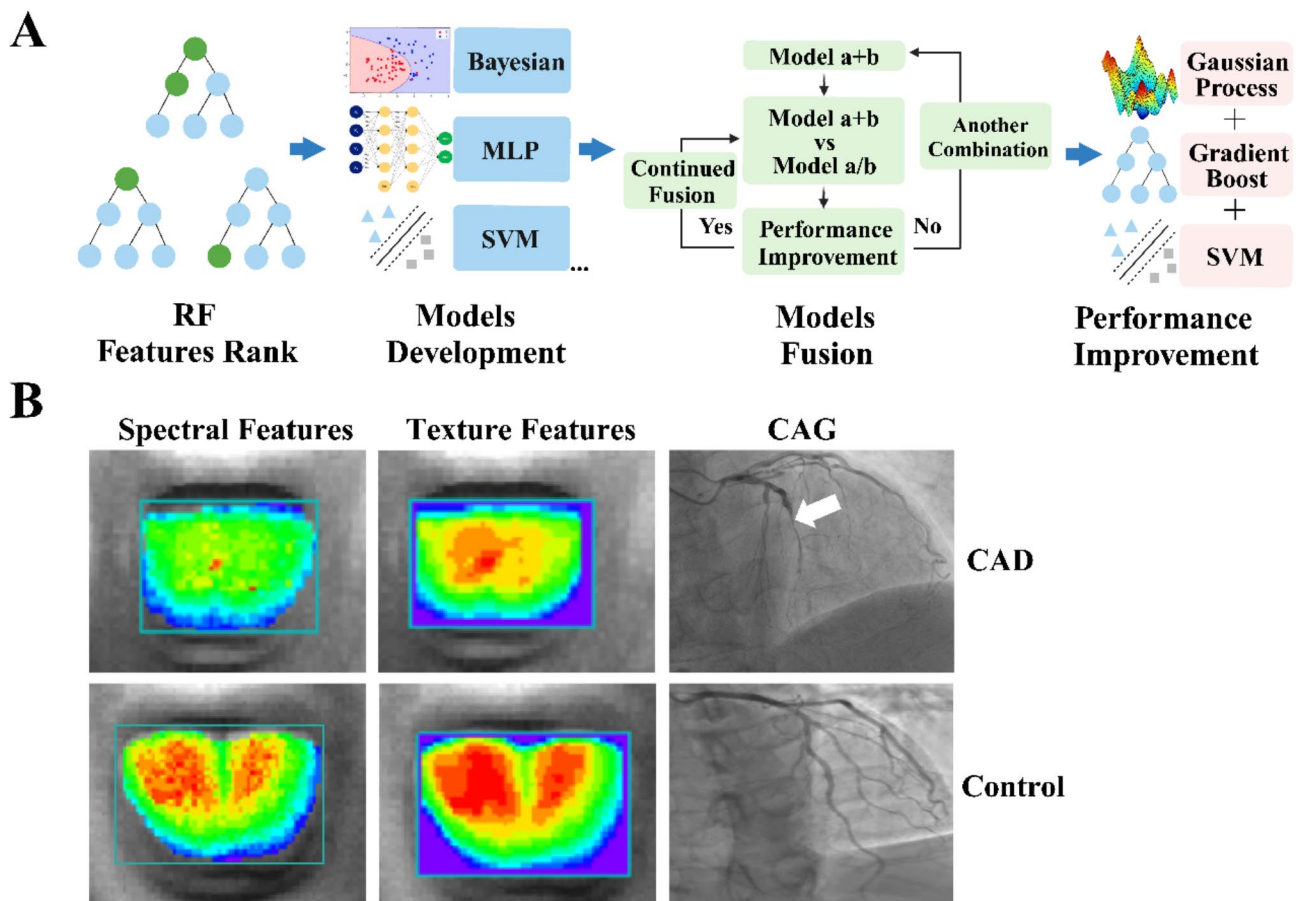
| Characteristics                                        | Training set<br>(n = 2555 images) | Internal set<br>(n = 1095 images) | External set<br>(n = 1095 images) | P-value |
|--------------------------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|---------|
| Age, median (IQR), years                               | 60 (53, 72)                       | 61 (56, 68)                       | 66 (60, 71)                       | 0.218   |
| Male, n (%)                                            | 64 (62.7%)                        | 24 (54.5%)                        | 32 (72.7%)                        | 0.208   |
| Height, median (IQR), cm                               | 169 (160, 171)                    | 165 (160, 172)                    | 168 (160, 173)                    | 0.853   |
| Weight, median (IQR), kg                               | 66 (58, 75)                       | 71 (59, 75)                       | 70 (65, 75)                       | 0.122   |
| Hypertension, n (%)                                    | 47 (46.1%)                        | 25 (56.8%)                        | 23 (52.3%)                        | 0.464   |
| Diabetes, n (%)                                        | 20 (19.6%)                        | 13 (29.5%)                        | 12 (27.3%)                        | 0.352   |
| Current or ever smokers, n (%)                         | 33 (32.4%)                        | 17 (38.6%)                        | 16 (36.4%)                        | 0.740   |
| History of alcohol, n (%)                              | 15 (14.7%)                        | 8 (18.2%)                         | 10 (22.7%)                        | 0.495   |
| Statin, n (%)                                          | 86 (84.3%)                        | 39 (88.6%)                        | 38 (86.4%)                        | 0.784   |
| β-blocker, n (%)                                       | 42 (41.2%)                        | 23 (52.3%)                        | 24 (54.5%)                        | 0.236   |
| RASS Inhibitors, n (%)                                 | 28 (27.5%)                        | 13 (29.5%)                        | 20 (45.5%)                        | 0.093   |
| CCB, n (%)                                             | 28 (27.5%)                        | 9 (20.5%)                         | 11 (25.0%)                        | 0.671   |
| Neutrophil, median (IQR), 10 <sup>9</sup>              | 3.69 (2.86, 4.86)                 | 3.40 (2.81, 4.54)                 | 4.21 (3.54, 5.23)                 | 0.202   |
| Lymphocyte, median (IQR), 10 <sup>9</sup>              | 1.51 (1.19, 2.06)                 | 1.68 (1.43, 1.97)                 | 1.46 (1.12, 1.80)                 | 0.251   |
| Platelet, median (IQR), 10 <sup>9</sup>                | 197 (169, 235)                    | 200 (166, 227)                    | 191 (159, 223)                    | 0.601   |
| CRP, median (IQR), mg/l                                | 0.72 (0.50, 2.00)                 | 0.95 (0.50, 2.53)                 | 0.71 (0.50, 2.95)                 | 0.530   |
| CK-MB, median (IQR), ng/ml                             | 0.35 (0.18, 0.67)                 | 0.35 (0.19, 0.62)                 | 0.67 (0.18, 2.02)                 | 0.100   |
| cTnI, median (IQR), ng/ml                              | 2.50 (0.01, 3.85)                 | 2.50 (0.01, 2.55)                 | 0.04 (0.01, 2.75)                 | 0.294   |
| Creatinine, median (IQR), μmol/l                       | 74 (61, 83)                       | 71 (61, 81)                       | 70 (66, 78)                       | 0.514   |
| eGFR, median (IQR), ml/min                             | 90 (77, 101)                      | 90 (78, 99)                       | 91 (90, 96)                       | 0.335   |
| Uric Acid, median (IQR), μmol/l                        | 348 (289, 413)                    | 360 (303, 423)                    | 339 (317, 372)                    | 0.475   |
| Blood Glucose, median (IQR), mmol/l                    | 5.64 (4.90, 7.07)                 | 5.57 (5.04, 5.91)                 | 5.90 (5.36, 7.07)                 | 0.086   |
| Triglyceride, median (IQR), mmol/l                     | 1.37 (1.04, 2.08)                 | 1.52 (1.26, 2.20)                 | 1.48 (1.15, 2.12)                 | 0.368   |
| Cholesterol, median (IQR), mmol/l                      | 4.07 (3.40, 4.62)                 | 4.31 (3.75, 5.06)                 | 3.85 (3.22, 4.39)                 | 0.101   |
| HDL-C, median (IQR), mmol/l                            | 1.06 (0.87, 1.26)                 | 1.16 (1.03, 1.32)                 | 1.05 (0.90, 1.32)                 | 0.314   |
| LDL-C, mean ± SD, mmol/l                               | 2.34 ± 0.72                       | 2.51 ± 0.90                       | 2.24 ± 0.98                       | 0.288   |
| Lipoprotein A, median (IQR), mg/l                      | 183 (97, 410)                     | 181 (103, 329)                    | 154 (112, 285)                    | 0.471   |
| NT-proBNP, median (IQR), pg/ml                         | 78 (36, 93)                       | 93 (43, 101)                      | 58 (36, 133)                      | 0.556   |
| LVEF, median (IQR), %                                  | 60.00 (60.00, 60.00)              | 60.00 (60.00, 60.00)              | 60.00 (59.50, 60.00)              | 0.923   |
| At least one coronary artery<br>with stenosis of ≥ 50% | 53 (52.0%)                        | 20 (45.5%)                        | 20 (45.5%)                        | 0.670   |

Data presented as mean ± standard deviation, median (IQR) or n (%). No data were missing. RASS, Renin-Angiotensin System; CCB, Calcium Channel Blocker; CRP, C-Reactive Protein; CK-MB, Creatine Kinase-MB; cTnI, Troponin I; eGFR, estimated Glomerular Filtration Rate; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; NT-proBNP, N-Terminal pro-B-Type Natriuretic Peptide; LVEF, Left Ventricular Ejection Fraction. P-value was obtained by comparison of the training, internal and external test sets

the risk of CAD in the internal test set (Fig. 5B). Additionally, the calibration plots of the GP-GB-SVM model exhibited the highest correlation between the probabilities of CAD and the actual observed outcomes in the external test set (Fig. 5E). The DCA for these six models demonstrated exceptional net benefit in the internal and external test sets across a range of threshold probabilities. The GP-GB-SVM model demonstrated a significantly higher net clinical benefit relative to the other five models, underscoring its practical utility in clinical decision-making processes (Fig. 5C and Fig. 5F). The corresponding ACC, SE, SP, F1 score, PPV, and NPV of these six models in internal test set were GP-GB-SVM (ACC:0.82, SE:0.70, SP:0.92, F1 score:0.78, PPV:0.88, NPV:0.79), GP (ACC:0.75, SE:0.55, SP:0.92, F1 score:0.67, PPV:0.85, NPV:0.71), GB (ACC:0.80, SE:0.88, SP:0.79, F1 score:0.74, PPV:0.81, NPV:0.79), SVM (ACC:0.70,

SE:0.76, SP:0.65, F1 score:0.71, PPV:0.67, NPV:0.75), RF (ACC:0.80, SE:0.68, SP:0.88, F1 score:0.74, PPV:0.81, NPV:0.79), TheilSen (ACC:0.66, SE:0.57, SP:0.74, F1 score:0.62, PPV:0.67, NPV:0.65) (Supplemental Table 1). The corresponding ACC, SE, SP, F1 score, PPV, and NPV of these six models in external test set were GP-GB-SVM (ACC:0.70, SE:0.90, SP:0.46, F1 score:0.75, PPV:0.60, NPV:0.90), GP (ACC:0.68, SE:0.74, SP:0.64, F1 score:0.70, PPV:0.67, NPV:0.70), GB (ACC:0.59, SE:0.68, SP:0.50, F1 score:0.63, PPV:0.58, NPV:0.61), SVM (ACC:0.52, SE:0.63, SP:0.40, F1 score:0.59, PPV:0.56, NPV:0.47), RF (ACC:0.66, SE:0.70, SP:0.62, F1 score:0.68, PPV:0.67, NPV:0.65), TheilSen (ACC:0.57, SE:0.61, SP:0.52, F1 score:0.60, PPV:0.58, NPV:0.55) (Supplemental Table 1). The remaining 25 models exhibit AUC ranges of 0.47 to 0.75 in the test set (Supplemental Table 1). The other





**Fig. 4** Model development and fusion. **A.** Overview of HSI-CAD models development. The part of “RF Features Rank” represented the significance of hyperspectral image features was assessed using random forests. The second part of “Models Development” displayed all features were individually incorporated into 30 different single machine learning algorithms, with each model incorporating 5 to 25 features. The third part of “Models Fusion” exhibited further enhancement in model performance was achieved through a decision level fusion strategy based on Area Under Curve (AUC) and F1 score. In the fourth part of “Performance Improvement”, we developed 30 HSI-CAD models and found that the combination of Gaussian Process, Gradient Boosting, and Support Vector Machine models exhibited the most substantial improvement in AUC and F1 score. **B.** Demonstration of hyperspectral features. Presentation of spectral and texture features for control and CAD patients MLP, Multilayer Perceptron; SVM, Support Vector Machine; RF, Random Forest; CAD, Coronary Artery Disease; CAG, Coronary Artery Angiography

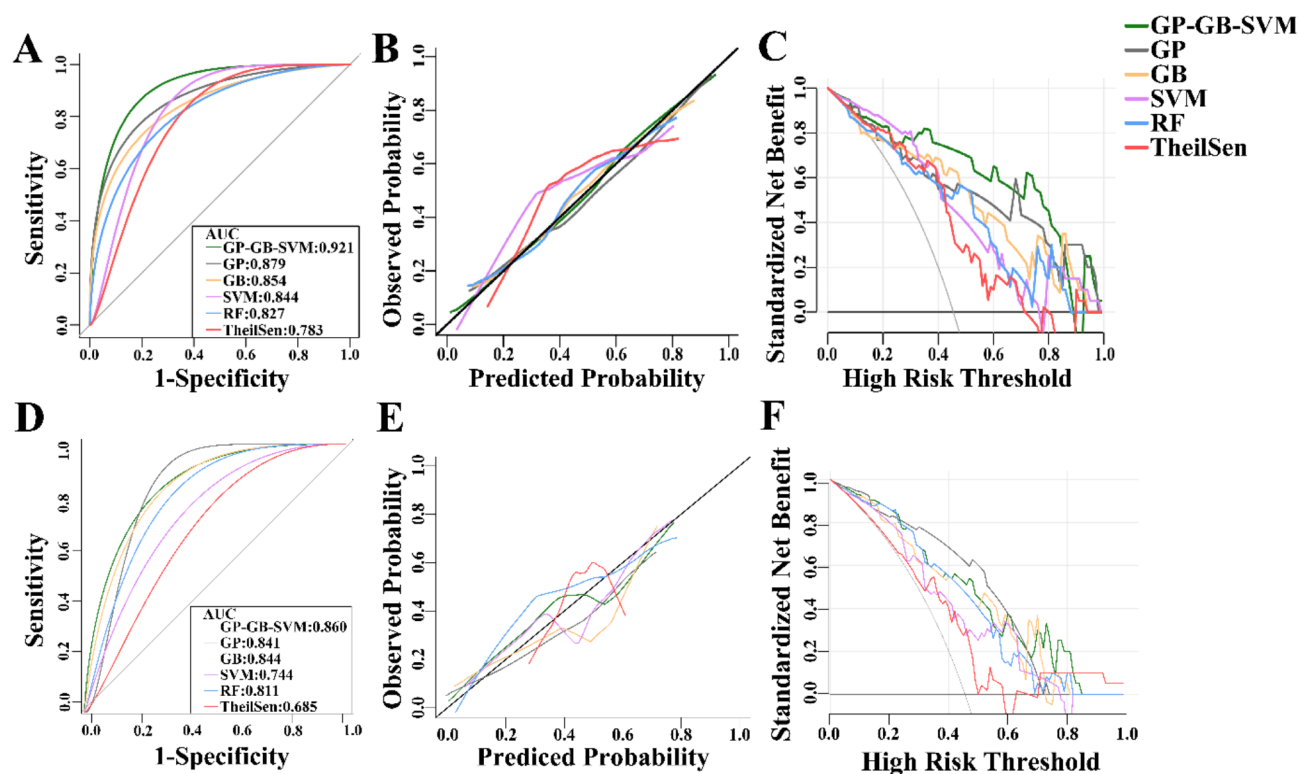
performance metrics were displayed in Supplemental Table 1 (Supplemental Table 1).

## Discussion

In this study, 16sRNA sequencing of dorsal tongue swabs revealed notable dysbiosis in the oral microbiota of CAD patients. The results showed significant differences in  $\alpha$  and  $\beta$  diversity between CAD and control patients. There were significant variations in relative abundance at phyla and genus levels between the two groups. Additionally, the reconstruction of microbial network was observed in CAD patients. 31 ML models based on hyperspectral images of the dorsal tongue (HSI-CAD model) were constructed to effectively predict CAD in patients presenting with chest pain. Among 31 prediction models assessed, the GP-GB-SVM fusion model demonstrated superior predictive performance, achieving an impressive AUC of 0.92 in the internal set and an AUC of 0.86 in the external

set. The findings advocated for oral microbiome dysbiosis in CAD patients and the implementation of this non-invasive screening technique, leveraging the remote sensing perspective to enhance the detection of CAD.

Pathophysiological, the dorsal surface of the tongue in individuals diagnosed with CAD exhibits notable characteristics, as revealed by our study. Previous studies have established a strong association between oral hygiene and the development of atherosclerotic cardiovascular disease (ASCVD) [24, 25]. The dysbiosis of the oral microbiota plays a significant role in the formation of atherosclerotic plaques and the progression of ASCVD, consequently increasing the risk of acute myocardial infarction [26]. The dorsal surface of the tongue harbors a substantial proportion of the oral microbiota, thus serving as an indicative measure for assessing the overall status of the oral microbiome [27]. In alignment with the previous findings, this study identified a more



**Fig. 5** Top 6 HSI-CAD models evaluation in the internal and external sets. **A.** ROC analysis in the internal set. The ROC curves of the top six HSI-CAD models for predicting CAD in the internal set. The AUC for the six models in the internal test set varied between 0.783 and 0.921 [GP-GB-SVM:0.921(0.839–1.000), GP:0.879(0.779–0.980), GB:0.854(0.737–0.972), SVM:0.844(0.728–0.959), RF:0.827(0.704–0.950), TheilSen:0.783(0.646–0.920)]. **B.** Calibration analysis in the internal set. In the internal test set, these models displayed a propensity to exaggerate the risk of developing CAD. **C.** Decision curve analysis in the internal set. DCA was performed on these top six HSI-CAD models to evaluate their predictive ability regarding coronary artery disease in the internal sets. At a range of threshold probabilities, the DCA of these six models shows superior net benefits in the internal test set. The GP-GB-SVM model exhibited the highest net clinical benefit. **D.** ROC analysis in the external test. The ROC curves of the top six HSI-CAD models for predicting CAD in the external set. The AUC values of the six models in the external test set range from 0.685 to 0.860 [GP-GB-SVM:0.860(0.754–0.967), GP:0.841(0.723–0.964), GB:0.844(0.728–0.959), SVM:0.744(0.596–0.892), RF:0.811(0.685–0.938), TheilSen:0.685(0.525–0.846)]. **E.** Calibration analysis in the external set. The calibration plots of the GP-GB-SVM model indicated that the highest correlation between the CAD probabilities and the actual observations in the external test set. **F.** Decision curve analysis in the external set. With a range of threshold probabilities, the DCA of these six models exhibited outstanding net benefits in the external test set. The net clinical benefit of the GP-GB-SVM model was more favorable in comparison to the other five models, emphasizing the utility of the model in the clinical decision-making process. GP-GB-SVM, Gaussian Process-Gradient Boosting-Support Vector-Machine; GP, Gaussian Process; GB, Gradient Boosting; SVM, Support Vector Machine; RF, Random Forest

pronounced dysbiosis of the oral microbiota in patients with CAD. Significant alterations in the oral microbiota composition were identified, particularly a decrease in species abundance and reduced network complexity and stability among patients with CAD. Previous studies have demonstrated that dysbiosis of the oral microbiota can facilitate the development of arterial plaques by augmenting immune-mediated inflammation and perturbing lipid metabolic homeostasis [28, 29]. For instance, *Fusobacterium* has been identified within the human arterial intima and is recognized for its role in exacerbating the progression of atherosclerosis through the promotion of inflammatory responses and the disruption of lipid metabolism [30]. The relative abundance of *Fusobacterium* was also significantly higher in CAD patients in this study. These insights suggest that changes in oral microbiota might be reflective or contributory to the

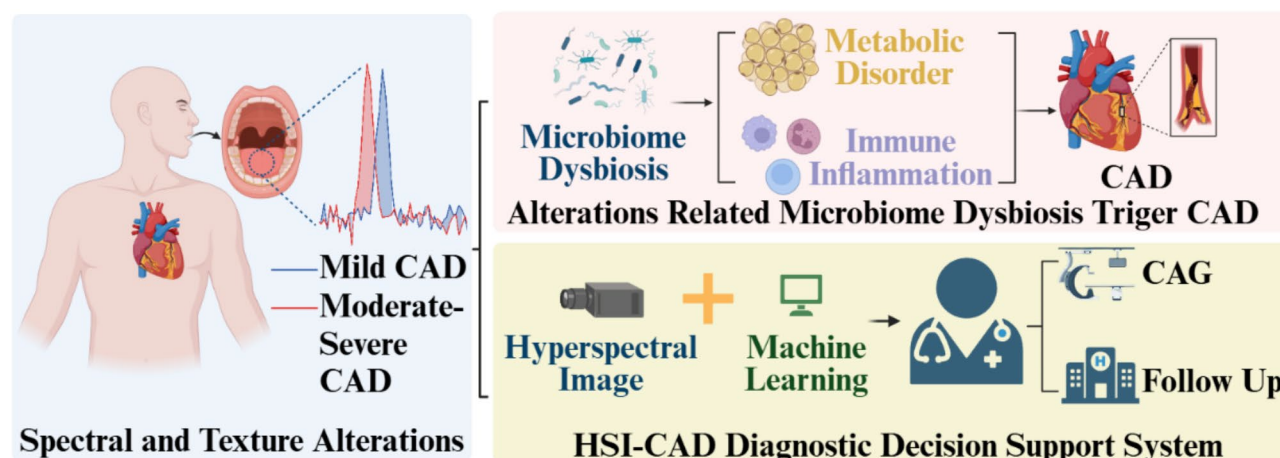
pathophysiological state of CAD, presenting a holistic view of how systemic diseases can manifest in oral environments. In addition, the results of the present study further elucidated the alterations in oral microbiota in patients with coronary artery disease compared with the results of previous studies. Firstly, we innovatively explored variations in oral microbiota in unstable angina pectoris patients. The present study was more focused on describing oral microbiota alterations in individuals at risk of coronary artery disease with symptoms of chest pain. The potential changes in oral microbiota in patients with different subtypes of coronary artery disease were further clarified. Secondly, compared with previous studies that concentrated only on the altered abundance of oral microbiota, the present study further found that the topological characteristics of oral microbial networks in patients with coronary artery disease, suggesting that an

imbalance of oral microenvironmental homeostasis may exist in the patients.

Clinically, the diagnosis of CAD is most accurately achieved through the utilization of coronary angiography, which is widely recognized as the gold standard in clinical practice. However, cardiac angiography is an invasive, intrusive test that carries a certain risk of surgical complications and imposes a certain financial burden on the patient [31]. If all patients with suspected coronary artery disease are subjected to angiography, it will result in a waste of medical resources. In low-resource settings characterized by limited funding and inadequate cardiovascular disease screening programs, there may be a scarcity of proficient medical practitioners or appropriate equipment. Therefore, the implementation of a straightforward and viable auxiliary diagnostic decision system could effectively assist healthcare professionals in primary medical institutions to promptly identify high-risk CAD patients. Previous research has identified significant phenotypic markers in patients with CAD. For instance, Christoffersen et al. demonstrated that male pattern baldness, diagonal earlobe crease, and xanthelasma significantly enhance the predictive accuracy of the Framingham model for cardiovascular disease risk [32]. In a separate study, the presence of a diagonal earlobe crease was identified as a predictor of CAD risk, demonstrating a sensitivity of 78% and a specificity of 61% [33]. Recently, several studies have confirmed that combining deep learning algorithms with facial RGB (Red Green Blue) or near-infrared images can effectively predict CAD [12, 34]. However, previous studies have not effectively addressed the confounding effects of variations in skin color across different ethnicities on superficial characteristics. Our study utilized hyperspectral images of the dorsal tongue to overcome these issues. Dorsal tongue images effectively avoid the interference of skin color differences in superficial characteristic information. Additionally, hyperspectral imaging, with its broad spectral range and high data dimensionality, provides more comprehensive superficial characteristic information for CAD patients. Furthermore, previous studies of coronary artery disease prediction models were based on demographic characteristics, clinical baseline information and relevant examination results for model construction [35, 36]. In contrast, the HSI-CAD model constructed in this study applied hyperspectral features of the participants for model development without other characteristics. On the one hand, a faster screening strategy for coronary artery disease was realized. On the other hand, the results suggested that specific spectral features may exist on the body surface of patients with coronary artery disease. These surface features indicate that patients may possess undetected risk factors. It provides new insights for the exploration of coronary artery disease related risk

factors. Additionally, this study innovatively combined oral microbiota and hyperspectral imaging to develop a novel diagnostic tool that utilizes noninvasive spectral image acquisition to explore potential alterations in patients' oral microbiota. The GP-GB-SVM model demonstrated good predictive performance (AUC: 0.92 in the internal set and AUC: 0.86 in the external set) without the need for baseline characteristics. The integration of HSI with the fusion model in our study facilitates the non-invasive, straightforward, and precise identification of high-risk CAD patients, particularly in resource-limited and remote regions. The GP-GB-SVM fusion model developed in this study is expected to play a role in assisting decision-making in coronary artery disease screening. In clinical practice, the GP-GB-SVM model combined with hyperspectral features can assist physicians in determining the risk of coronary artery disease and the need for further coronary angiography in high-risk patients with chest pain. With the advantages of short time-consumption and high accuracy of the fusion model, it is expected to realize the accurate identification of high-risk patients and the exclusion of low-risk patients to reduce unnecessary examinations. In addition, some high-risk patients with chest pain may be initially evaluated in the primary care setting. The fusion model can be used to screen for the risk of coronary artery disease in these patients, and can be used to assist in early diagnosis and treatment. Furthermore, the fusion model can be used to determine the risk in patients who present with symptoms suspected to be coronary artery disease at home. The fusion model can be used to assist patients in deciding whether they need to attend hospitals to complete the relevant examinations and treatments. Therefore, this fusion model was expected to realize effective assistance in the diagnosis of CAD and decision-making of treatment strategies. And it also further confirmed the significant potential application value of artificial intelligence in CAD diagnosis and treatment [37].

Overall, this study elucidated the significant changes in oral microbiota in patients with coronary artery disease, and combined hyperspectral imaging and machine learning algorithms to screen the patients' oral characteristic spectra. On the one hand, the microbial changes in the oral cavity suggested that despite the traditional risk factors, such as diabetes and hyperlipidemia, the microbial alterations in the oral cavity may also be an important factor in the development of coronary artery disease. These patients may require not only control of traditional risk factors, but also attention to oral hygiene and health, with intervention as necessary. On the other hand, the presence of specific spectra in the oral cavity suggests microenvironmental alterations that may be caused by oral microbial and human interactions. Coronary heart disease is a disease in combination with multiple risk



**Fig. 6** Oral alterations in CAD patients and prospects for application of HSI-CAD models for CAD screening

factors, so it is meaningful to integrate multiple risk factors to determine the degree of risk of coronary heart disease. The characteristic spectra in this study represent microenvironmental patterns that may be the result of a combination of different risk factors, such as microbial changes and metabolic variations in the human body. Therefore, these feature spectra may become new clinical biomarkers for identifying patients with coronary heart disease. However, further inclusion of more patients with different types of coronary artery disease is still needed in the future to increase the certainty of this evidence. Inclusion of more patients with different types of coronary artery disease would help to further identify and clarify whether the oral microbiota of the coronary artery disease population experience changes with disease progression. Additionally, the inclusion of more patient data can be used to further optimize the HSI-CAD model and improve its clinical value. Therefore, further popularization of sequencing and hyperspectral imaging is required to promote the use of this approach in clinical practice. These technologies need to be further generalized to resource-limited areas. Secondly, more populations from different regions need to be included to clarify whether these technologies are affected by regional and population differences.

## Conclusion

In conclusion, oral hygiene is closely related to cardiovascular disease. Tongue microbiota dysbiosis can be observed in CAD patients. Therefore, further studies may need to concentrate on the changes in oral microbiota in CAD patients to clarify the exact mechanism between these alterations and CAD. The integration of remote sensing technology introduces a novel dimension to medical diagnostics, significantly enhancing the potential for early detection of CAD through non-invasive methodologies (Fig. 6).

## Limitations of the study

However, there are still some limitations in this study. Firstly, oral microbiome analysis was only described in the patients and did not further explore the exact mechanism or potential biomarker of the special microbiome on CAD. Secondly, the study's sample size and population diversity may be limited, potentially impacting the generalizability of the results. It is necessary to include larger and more diverse multi-ethnic cohorts to validate the findings across different demographic groups effectively. Thirdly, this study only applied machine learning algorithms for feature screening and model construction. Therefore, algorithm innovations still need to be further explored, including algorithm optimization, model efficiency enhancement, model interpretation, data processing optimization, etc. In addition, the models in this study were not constructed to incorporate baseline characteristics, and further refinement of the models by combining multiple characteristics is needed in the future. Finally, relying solely on advanced HSI technology may restrict the broader applicability of this diagnostic approach, particularly in resource-limited settings where such technology is not readily accessible.

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12872-025-04555-5>.

**Supplementary Material 1: Supplemental Fig. 1.** LEfSe analysis of oral microbiota. **Supplemental Fig. 2.** Study flowchart. **Supplemental Table 1.** Performance evaluation of HSI-CAD models in the internal and external test

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Not applicable.

## Author contributions

L.Y., Y.Z., X.W. and J.Y. had primary responsibility for the study design and data curation Z.L., X.Y., D.Z. and X.S. performed the data analysis, the models



establishment and drafted the manuscript. L.L., T.X., X.L., F.Z. and W.L. participated in data collection. L.Y., Y.Z., X.W. and S.W. have accessed and verified the data. Z.L., X.Y., D.Z. and X.S. contributed equally to this work. All authors read and approved the final manuscript.

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### Data availability

The databases underlying the results in the current study can be obtained after de-identification from the corresponding author (e-mail lileiyu@whu.edu.cn) upon reasonable request.

### Declarations

#### Ethics approval and consent to participate

The research protocol and consent processes received approval from the Institutional Review Board of Renmin Hospital of Wuhan University (WDRY2023-K084, chict.org.cn identifiers ChiCTR2300073683, Registration Date:2023-7-19) and Yichang Central People's Hospital (No.2024-216-01). The procedures in this study adhered to the tenets of the Declaration of Helsinki.

#### Consent for publication

All eligible participants signed informed consent for study involvement, including voluntarism, information disclosure, decision-making capacity, and use of their hyperspectral images and swab of tongue in research before the CAG procedure.

#### Competing interests

The authors declare no competing interests.

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### References

1. Mensah GA, Fuster V, Murray CJL, Roth GA. Global Burden of Cardiovascular diseases, risks collaborators. Global burden of cardiovascular diseases and risks, 1990–2022 [J]. *J Am Coll Cardiol*. 2023;82(25):2350–473. <https://doi.org/10.1016/j.jacc.2023.11.007>.
2. Tonelli A, Lumngwena EN, Ntusi NA. B. The oral microbiome in the pathophysiology of cardiovascular disease [J]. *Nat Rev Cardiol*. 2023;20(6):386–403. <https://doi.org/10.1038/s41569-022-00825-3>.
3. Park SY, Kim SH, Kang SH, Yoon CH, Lee HJ, Yun PY, Youn TJ, Chae IH. Improved oral hygiene care attenuates the cardiovascular risk of oral health disease: a population-based study from Korea [J]. *Eur Heart J*. 2019;40(14):1138–45. <https://doi.org/10.1093/eurheartj/ehy836>.
4. Zhuang Z, Gao M, Lv J, Yu C, Guo Y, Bian Z, Yang L, Du H, Chen Y, Ning F, Liu H, Chen J, Chen Z, Huang T, Li L. Associations of toothbrushing behaviour with risks of vascular and nonvascular diseases in Chinese adults [J]. *Eur J Clin Invest*. 2021;51(12):e13634. <https://doi.org/10.1111/eci.13634>.
5. Li Y, Liu Y, Cui J, Zhu M, Wang W, Chen K, Huang L, Liu Y. Oral-gut microbial transmission promotes diabetic coronary heart disease [J]. *Cardiovasc Diabetol*. 2024;23(1):123. <https://doi.org/10.1186/s12933-024-02217-y>.
6. Koren O, Spor A, Felin J, Fåk F, Stombaugh J, Tremaroli V, Behre CJ, Knight R, Fagerberg B, Ley RE, Bäckhed F. Human oral, gut, and plaque microbiota in patients with atherosclerosis [J]. *Proc Natl Acad Sci U S A*. 2011;108(Suppl 1):4592–8. <https://doi.org/10.1073/pnas.1011383107>.
7. Radaic A, Kapila YL. The oralome and its dysbiosis: new insights into oral microbiome-host interactions [J]. *Comput Struct Biotechnol J*. 2021;19:1335–60. <https://doi.org/10.1016/j.csbj.2021.02.010>.
8. Schenkein HA, Papapanou PN, Genco R, Sanz M. Mechanisms underlying the association between periodontitis and atherosclerotic disease [J]. *Periodontol*. 2020;83(1):90–106. <https://doi.org/10.1111/prd.12304>.
9. Wilbert SA, Mark Welch JL, Borisy GG. Cell Rep. 2020;30(12):4003–e40154003. <https://doi.org/10.1016/j.celrep.2020.02.097>. Spatial ecology of the human tongue dorsum microbiome [J].
10. Christoffersen M, Frikke-Schmidt R, Schnohr P, Jensen GB, Nordestgaard BG, Tybjaerg-Hansen A. Visible age-related signs and risk of ischemic heart disease in the general population: a prospective cohort study [J]. *Circulation*. 2014;129(9):990–8. <https://doi.org/10.1161/circulationaha.113.001696>.
11. Ang M, Wong W, Park J, Wu R, Lavanya R, Zheng Y, Cajucom-Uy H, Tai ES, Wong T. Y. corneal arcus is a sign of cardiovascular disease, even in low-risk persons [J]. *Am J Ophthalmol*. 2011;152(5):864–e871861. <https://doi.org/10.1016/j.ajo.2011.04.014>.
12. Lin S, Li Z, Fu B, Chen S, Li X, Wang Y, Wang X, Lv B, Xu B, Song X, Zhang YJ, Cheng X, Huang W, Pu J, Zhang Q, Xia Y, Du B, Ji X, Zheng Z. Feasibility of using deep learning to detect coronary artery disease based on facial photo [J]. *Eur Heart J*. 2020;41(46):4400–11. <https://doi.org/10.1093/eurheartj/ehaa640>.
13. Zeng M, Smith L, Bird A, Trinh VQ, Bacchi S, Harvey J, Jenkinson M, Scroop R, Kleinig T, Jannes J, Palmer L. J. Predictions for functional outcome and mortality in acute ischaemic stroke following successful endovascular thrombectomy [J]. *BMJ Neurol Open*. 2024;6(1):e000707. <https://doi.org/10.1136/bmjn-2024-000707>.
14. Yako Motoki Y, Yoshikazu K, Takayuki H, Chikaki N, Akihiro T, Klaas S, Nick H, Taku, Ishikawa Atsushi. Video-rate hyperspectral camera based on a cmos-compatible random array of fabry-pérot filters [J]. *Nat Photonics*. 2023;17(3):218–23. <https://doi.org/10.1038/s41566-022-01141-5>.
15. Clancy NT, Jones G, Maier-Hein L, Elson DS, Stoyanov D. Surgical spectral imaging [J]. *Med Image Anal*. 2020;63:101699. <https://doi.org/10.1016/j.medi.2020.101699>.
16. Leon R, Fabelo H, Ortega S, Cruz-Guerrero IA, Campos-Delgado DU, Szolna A, Pineiro JF, Espino C, O'Shanahan AJ, Hernandez M, Carrera D, Bisshopp S, Sosa C, Balea-Fernandez FJ, Morera J, Clavo B, Callico GM. Hyperspectral imaging benchmark based on machine learning for intraoperative brain tumour detection [J]. *NPJ Precis Oncol*. 2023;7(1):119. <https://doi.org/10.1038/s41698-023-00475-9>.
17. Hadoux X, Hui F, Lim JKH, Masters CL, Pebay A, Chevalier S, Ha J, Loi S, Fowler CJ, Rowe C, Villemagne VL, Taylor EN, Fluke C, Soucy JP, Lesage F, Sylvestre JP, Rosa-Neto P, Mathotaarachchi S, Gauthier S, Nasreddine ZS, Arbour JD, Rheaume MA, Beaulieu S, Dirani M, Nguyen CTO, Bui BV, Williamson R, Crowston JG, van Wijngaarden P. Non-invasive in vivo hyperspectral imaging of the retina for potential biomarker use in alzheimer's disease [J]. *Nat Commun*. 2019;10(1):4227. <https://doi.org/10.1038/s41467-019-12242-1>.
18. Kazune S, Caica A, Volceka K, Suba O, Rubins U, Grabovskis A. Relationship of mottling score, skin microcirculatory perfusion indices and biomarkers of endothelial dysfunction in patients with septic shock: an observational study [J]. *Crit Care*. 2019;23(1):311. <https://doi.org/10.1186/s13054-019-2589-0>.
19. Yoon J, Joseph J, Waterhouse DJ, Luthman AS, Gordon GSD, di Pietro M, Januszewicz W, Fitzgerald RC, Bohndiek S. E. A clinically translatable

- hyperspectral endoscopy (hyse) system for imaging the gastrointestinal tract [J]. *Nat Commun*. 2019; 10(1): 1902. <https://doi.org/10.1038/s41467-019-0948-4-4>
20. Greenman Robert L, Svetlana P, Xiaoen W, Lyons Thomas E, Thanh D, Lydia L, Jenny GJMF, Lalita K. Veves Aristidis. Early changes in the skin microcirculation and muscle metabolism of the diabetic foot [J]. *Lancet*. 2005;366(9498):1711–7. [https://doi.org/10.1016/s0140-6736\(05\)67696-9](https://doi.org/10.1016/s0140-6736(05)67696-9).
21. Chen H, Li Q, Li M, Liu S, Yao C, Wang Z, Zhao Z, Liu P, Yang F, Li X, Wang J, Zeng Y, Tong X. Microbial characteristics across different tongue coating types in a healthy population [J]. *J Oral Microbiol*. 2021;13(1):1946316. <https://doi.org/10.1080/20002297.2021.1946316>.
22. Martinez B, Leon R, Fabelo H, Ortega S, Pineiro JF, Szolna A, Hernandez M, Espino C, Shanahan AJ, Carrera D, Bishopp S, Sosa C, Marquez M, Camacho R, Plaza ML, Morera J. Callico. Most relevant spectral bands identification for brain cancer detection using hyperspectral imaging [J]. *Sens (Basel)*. 2019;19(24). <https://doi.org/10.3390/s19245481>.
23. Jiangping S, Zhe Z, Wei W, Yunhu S, Jie H, Hongyue W, Hong Z, Shengshou H. Assessment of coronary artery stenosis by coronary angiography: a head-to-head comparison with pathological coronary artery anatomy [J]. *Circ Cardiovasc Interv*. 2013;6(3):262–8. <https://doi.org/10.1161/CIRCINTERVENTIONS.112.000205>.
24. F DeStefano RF, Anda HS, Kahn DF, Williamson CM, Russell. Dental disease and risk of coronary heart disease and mortality [J]. *BMJ*. 1993; 13;306(6879): 688–691. <https://doi.org/10.1136/bmj.306.6879.688>
25. Dietrich T, Sharma P, Walter C, Weston P, Beck J. The epidemiological evidence behind the association between periodontitis and incident atherosclerotic cardiovascular disease [J]. *J Periodontol*. 2013;84(4 Suppl):S70–84. <https://doi.org/10.1902/jop.2013.134008>.
26. Mattila KJ, Nieminen MS, Valtanen VV, Rasi VP, Kesäniemi YA, Syrjälä SL, Jungell PS, Isoluoma M, Hietaniemi K, Jokinen MJ. Association between dental health and acute myocardial infarction [J]. *BMJ*. 1989;298(6676):779–81. <https://doi.org/10.1136/bmj.298.6676.779>.
27. Deo PN, Deshmukh R. Oral microbiome: unveiling the fundamentals [J]. *J Oral Maxillofac Pathol*. 2019;23(1):122–8. [https://doi.org/10.4103/jomfp.JOMFP\\_304\\_18](https://doi.org/10.4103/jomfp.JOMFP_304_18).
28. Chandy Swaroop. Evaluation of c-reactive protein and fibrinogen in patients with chronic and aggressive periodontitis: a clinico-biochemical study [J]. *J Clin Diagn Res*. 2017. <https://doi.org/10.7860/jcdr/2017/23100.9552>.
29. Ozlem Fentoglu, Bozkurt F, Yesim. The bi-directional relationship between periodontal disease and hyperlipidemia [J]. *Eur J Dent*. 2008;2(2):142–6.
30. Ford PJ, Gemmell E, Hamlet SM, Hasan A, Walker PJ, West MJ, Cullinan MP, Seymour GJ. Cross-reactivity of groel antibodies with human heat shock protein 60 and quantification of pathogens in atherosclerosis [J]. *Oral Microbiol Immunol*. 2005;20(5):296–302.
31. Vrints C, Andreotti F, Koskinas KC, Rossello X, Adamo M, Ainslie J, Banning AP, Budaj A, Buechel RR, Chiariello GA, Chieffo A, Christodorescu RM, Deaton C, Doenst T, Jones HW, Kunadian V, Mehili J, Milojevic M, Piek JJ, Pugliese F, Rubboli A, Semb AG, Senior R, Ten Berg JM, Van Belle E, Van Craenenbroeck EM, Vidal-Perez R, Winther S. 2024 esc guidelines for the management of chronic coronary syndromes [J]. *Eur Heart J*. 2024;45(36):3415–537. <https://doi.org/10.1093/eurheartj/ehae177>.
32. Christoffersen M, Tybjaerg-Hansen A. Visible aging signs as risk markers for ischemic heart disease: epidemiology, pathogenesis and clinical implications [J]. *Ageing Res Rev*. 2016;25:24–41. <https://doi.org/10.1016/j.arr.2015.11.002>.
33. Wang Y, Mao LH, Jia EZ, Li ZY, Ding XQ, Ge PC, Liu Z, Zhu TB, Wang LS, Li CJ, Ma WZ, Yang ZJ. Relationship between diagonal earlobe creases and coronary artery disease as determined via angiography [J]. *BMJ Open*. 2016;6(2):e008558. <https://doi.org/10.1136/bmjopen-2015-008558>.
34. Kung M, Zeng J, Lin S, Yu X, Liu C, Shi M, Sun R, Yuan S, Lian X, Su X, Zhao Y, Zheng Z, Ji X. Prediction of coronary artery disease based on facial temperature information captured by non-contact infrared thermography [J]. *BMJ Health Care Inf*. 2024;31(1). <https://doi.org/10.1136/bmjhci-2023-100942>.
35. Forrester IS, Petrazzini BO, Duffy A, Park JK, Marquez-Luna C, Jordan DM, Rocheleau G, Cho JH, Rosenson RS, Narula J, Nadkarni GN, Do R. Machine learning-based marker for coronary artery disease: derivation and validation in two longitudinal cohorts [J]. *Lancet*. 2023;401(10372):215–25. [https://doi.org/10.1016/S0140-6736\(22\)02079-7](https://doi.org/10.1016/S0140-6736(22)02079-7).
36. Wei D, Melgarejo JD, Van Aelst L, Vanassche T, Verhamme P, Janssens S, Peter K, Zhang ZY. Prediction of coronary artery disease using urinary proteomics [J]. *Eur J Prev Cardiol*. 2023;30(14):1537–46. <https://doi.org/10.1093/eurjpc/zwad087>.
37. Aşkın Lütfü P, Esra Hoşoğlu, Yusuf, Tanrıverdi Okan. The application of artificial intelligence in the field of cardiovascular diseases focuses on both diagnostic and therapeutic aspects [J]. *Experimental Appl Med Sci*. 2024;5(2):22–35. <http://doi.org/10.46871/eams.1438927>.

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